OLANZAPINE

Weight gain & diabetes

Low EPS

CHEMICAL NAME 2-methyl-4-(4-methylpiperazin-1-yl)-10H-thieno[2,3-b][1,5]benzodiazepine
CLASS Second-generation (atypical) antipsychotic (thienobenzodiazepine)
TRADE NAMES Zyprexa · Zyprexa Velotab · ZypAdhera (depot) · Olapin

THERAPEUTIC USES / INDICATIONS

- Schizophrenia.
- Bipolar mania and prevention of recurrence.
- Acute agitation (IM).
- Chemotherapy-induced nausea (off-label, 5–10 mg); treatment-resistant depression (with fluoxetine).

ADMINISTRATION

- **PO** once daily, with or without food; orodispersible tablet on tongue.
- **IM** short-acting for agitation (not within 1 h of IM benzodiazepine).
- **Depot (pamoate):** observe 3 h after injection (post-injection delirium/sedation syndrome).

DOSAGES

- **Schizophrenia:** 10 mg daily; range 5–20 mg.
- **Mania:** 15 mg daily (10 mg with mood stabiliser).
- **Agitation IM:** 5–10 mg (max 20 mg/day).
- **Elderly/smokers differ:** start 5 mg in elderly; smokers may need higher doses.

PHARMACOKINETICS

- **Peak:** 5–8 h (PO); IM 15–45 min.
- Metabolised by CYP1A2 (↑ by smoking) and glucuronidation.
- **t½:** 21–54 h.

NURSING MANAGEMENT

- Metabolic monitoring: weight, BMI, waist, BP, glucose/HbA1c, lipids at baseline, 3 months, then yearly.
- Assess sedation, mental state, EPS.
- Depot: 3-h observation.
- Smoking changes → dose review.

PHARMACODYNAMICS

- Antagonist at **D2 and 5-HT2A** receptors (atypical profile → fewer EPS); also H1, M, α1 → sedation, weight gain, anticholinergic effects.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Weight gain, increased appetite, sedation.
- Dry mouth, constipation, dizziness.
- Raised prolactin (mild).

ADVERSE EFFECTS (SERIOUS)

- Metabolic syndrome: diabetes, dyslipidaemia.
- NMS; tardive dyskinesia (lower risk).
- ↑ stroke/mortality in elderly with dementia.
- Neutropenia; DRESS.

CONTRAINDICATIONS & CAUTIONS

- Known risk of angle-closure glaucoma.
- **Caution:** diabetes, obesity, elderly with dementia, prostatic hypertrophy, seizures.

KEY INTERACTIONS

- **Smoking, carbamazepine:** ↓ levels.
- **Fluvoxamine, ciprofloxacin:** ↑ levels.
- **Benzodiazepines (IM):** cardiorespiratory depression.
- **Alcohol:** ↑ sedation.

PATIENT EDUCATION

- Healthy diet and exercise to limit weight gain.
- Report excessive thirst, urination (diabetes), fever with stiffness.
- Tell your team if you stop or start smoking.
- Avoid alcohol.

KEY POINTS

- Atypical; D2/5-HT2A antagonist.
- Weight gain, diabetes, dyslipidaemia.
- 10 mg daily for schizophrenia.
- Smoking lowers levels (CYP1A2).

SOURCES CHECKED **[805]** Olanzapine 10 mg orodispersible tablets (UK SmPC) **[806]** Olanzapine, oral (US label) **[807]** Olanzapine, oral (US label, Modavar Pharmaceuticals LLC)
Full references p. 432–457